Scope and Objective

Surface electromyography (sEMG)-based pattern recognition has emerged as a primary paradigm for intuitive multi-gesture control in upper-limb prostheses. While high classification accuracies are frequently demonstrated on healthy intact populations, translating these models to clinical amputee populations presents substantial challenges.

By restricting the scope exclusively to NinaPro DB3—a benchmark dataset comprising 11 transradial amputees, 12 sEMG channel streams, and up to 50 distinct gesture classes—this evaluation directly addresses the primary real-world barriers: anatomical disruption, localized muscle atrophy, variable SNR, and high inter-subject heterogeneity

Related Work Summary

Evaluations conducted on NinaPro DB3 consistently reveal a pronounced performance discrepancy when compared against intact benchmark datasets (e.g., NinaPro DB2). Recent methodological advancements have concentrated on three main domain paradigms:

- **Classical Machine Learning Baselines:** Hand-crafted time-domain (TD) features combined with Support Vector Machines (SVM) or Random Forests (RF) yield subpar performance (<50% accuracy on 50 classes) due to non-stationarity and degraded signal-to-noise ratios.
- **Advanced Feature Projections:** Shift-invariant representations, such as log-scaled time-dependent power spectral density moments (TD-PSD2), effectively mitigate cross-subject non-stationarity and muscle structure alterations.
- **Transfer Learning & Domain Adaptation:** Architectures utilizing Gradient Reversal Layers (GRL), Triplet Loss, or multi-scale state-space models achieve fast fine-tuning on unseen amputees with minimal calibration overhead.

State-of-the-Art Literature Benchmark

Study & Citation	Target Classes	Processing & Feature Pipeline	Model Architecture	DB3 Performance	Key Findings / Clinical Contribution
Zhai et al. (2017)	10 classes*	Spectrogram representation / 200 ms window	Self-recalibrating 2D-CNN	73.31% Acc	Implemented self-recalibration mechanisms to counteract short-term muscle fatigue in transradial limbs.
MSDS-FusionNet (2026)	49 classes	Dual-stream multi-scale temporal & frequency maps	Multi-Scale Mamba (MSM) + BAFM	72.32% Acc	Captured both short- and long-term state-space temporal dynamics on intact and amputee cohorts.

Study & Citation	Target Classes	Processing & Feature Pipeline	Model Architecture	DB3 Performance	Key Findings / Clinical Contribution
Pancholi et al. (2021)	49 classes	Time-dependent power spectrum moments (TD-PSD1) / 150 ms	DLPR / Deep CNN	81.67% Acc	Proved that spectral moment representations significantly lower cross-subject variance across amputees.
Sandoval-Espino et al. (2022)	50 classes	Log-scaled spectral moments (TD-PSD2) + 'Mix Image'	8-Layer GengNet CNN	90.23% Acc	Achieved state-of-the-art accuracy on DB3 while maintaining real-time latency thresholds (<150 ms).
Fratti et al. (2024)	14 ADL classes	Raw sEMG / 150 ms window with circular channel padding	MSCNN + Transfer Learning (GRL / Triplet)	97.00% F1	Optimized for practical daily-living gestures; achieved rapid user fine-tuning in ~1.4 epochs.

Critical Research Gap Analysis

Despite notable classification milestones achieved on offline benchmark databases, several systemic gaps continue to prevent the successful clinical translation of deep learning-driven sEMG controllers to upper-limb amputees:

Anatomical Heterogeneity and Variable Forearm Stump Geometry

- **Identified Issue:** The 11 transradial amputee participants in NinaPro DB3 possess distinct levels of residual limb length, amputation etiology, residual muscle mass, and phantom limb sensation capabilities.
- **Unresolved Research Gap:** Existing deep neural networks fail to achieve acceptable zero-shot cross-subject generalization. There is a need for lightweight, subject-adaptive domain alignment techniques (e.g., contrastive self-supervised pre-training) that personalize models using minimal daily calibration trials.

Signal Degradation, Low SNR, and Severe Muscle Atrophy

- **Identified Issue:** Residual amputee muscle tissue generates sEMG signals characterized by significantly reduced Signal-to-Noise Ratios (SNR), unpredictable co-activations, and spatial crosstalk relative to intact control subjects.
- **Unresolved Research Gap:** Directly feeding raw sEMG into deep CNN/Transformer models often induces severe overfitting to background noise artifacts. Robust preprocessing pipelines combining shift-invariant log-spectral transformations (such as TD-PSD2) with structural channel topology mapping remain essential.

Granular Gesture Pool vs. Clinical Amputee Stamina

- **Identified Issue:** NinaPro DB3 evaluates up to 50 complex gesture classes (fine finger, wrist, and grasp patterns). Executing full repetition protocols induces acute muscle fatigue and localized pain in amputee subjects.
- **Unresolved Research Gap:** A critical discrepancy exists between evaluating comprehensive gesture pools for benchmark accuracy versus deploying clinically viable controllers restricted to high-confidence Activities of Daily Living (ADL) gesture sets (e.g., 10–14 classes) with continuous, online self-recalibration mechanisms.